PROCESS OR MATERIALS REVIEW REQUEST FORM

SNF Stanford Nanofabrication Facility

Request Title: ALD of WO3 in Fiji 3

Requestor: Dave Palm **Date: 8/30/2018**

Non-SNF staff, please complete the following:

Badger ID: dpalm Email: dpalm@stanford.edu

Company name/PI: Prof. Thomas Jaramillo

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups and the Stanford Chemical Safety Data Base sections on this website to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

   - precursor: bis(*tert*-butylimino)bis(dimethylamino)tungsten(VI) (BTBMW, CAS: 406462-43-9, ((CH3)3CN)2W(N(CH3)2)2,); Storage Group B; corrosive, flammable liquid (Class I-B), water reactive (Class 1)
   - product: WO3 (tungsten trioxide)

2. **Vendor/manufacturer info:** address and phone number, website URL.

   - SNF already purchases this chemical from Strem (https://www.strem.com/catalog/v/74-1225/78/tungsten_406462-43-9)
   - 7 Mulliken Way, Newburyport, MA 01950-4098 USA, Tel: (978) 499 1600

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

   - In speaking with Dr. Michelle Rincon, we believe that the Ar-boosted precursor delivery system available only on the Fiji 3 ALD system is the best option for getting repeatable deposition of WO3 from this precursor, which has relatively low vapor pressure.
   - A recent publication employs this same strategy (as in attached article clipping from Zhang, T., Lee, C., Gong, B., & Hoex, B. (2018). Thermal stability analysis of WOx and MoOx as hole-selective contacts for Si solar cells using in situ XPS (Vol. 040027, p. 040027).

http://doi.org/10.1063/1.5049290):

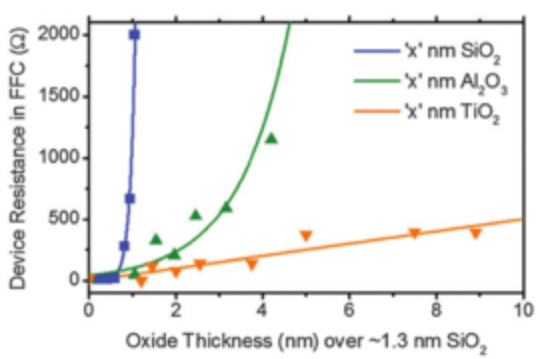
EXPERIMENT

Sample Preparation

4 Ω cm p-type float-zone (FZ) c-Si wafers were used as a substrate in this work. Prior to deposition, all wafers went through a standard RCA (Radio Corporation of America) cleaning process and were dipped in 2% w/v HF solution to remove the native silicon oxide. The WO_x films were deposited from bis(tertbutylimino) bis(dimethylamino)tungsten(VI) [$(NtBu)_2(NMe_2)_2W$, 97%, Strem Chemicals] and an O_2 plasma using atomic layer deposition (ALD, Fiji G2 Cambridge Nanotech). The precursor was heated to 80 °C and delivered to the ALD chamber by a booster system due to the low vapour pressure of the W-precursor. Ar gas was used both as precursor carrier and purge gas. The substrate temperature was 200 °C during the ALD process. The plasma power was 300 W and the exposure time was 20 s. The MoO_x films were deposited by thermally evaporating molybdenum (VI) oxide

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that f the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

- The atomic layer deposition will be carried out in Fiji 3, an ALD tool classified as gold-contaminated. After the deposition, 125 cycles of SiO2 ALD (approx. 100 A at 0.8 A/cycle) will be deposited in order to avoid cross-contamination. (Note, this thickness of SiO2 will need to be verified by experiments)
- According to work from Prof Paul McIntyre's group, SiO2 can serve as a resistive sealant in solid-state electrochemical devices, as shown in the plot below (from Scheuermann, A. G., Kemp, K. W., Tang, K., Lu, D. Q., Satterthwaite, P. F., Ito, T., … McIntyre, P. C. (2016). Conductance and capacitance of bilayer protective oxides for silicon water splitting anodes. Energy & Environmental Science, 9(2), 504–516. http://doi.org/10.1039/C5EE02484F):



- To verify the effectiveness of the SiO2 blocking layer at high temperature, we will check the tungsten contamination level on silicon wafers using VPD-ICPMS.
- We will clean 3 wafers, then keep one as a control, cycle one through the chamber after the

WO3 deposition, then one after the SiO2 deposition. All wafers will be put face down on the sample plate. We will then judge if 125 cycles of ALD SiO2 is sufficient for blocking tungsten contamination or not.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

   - The precursor is already used in SNF. It is purchased in 5 g bottles.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

   - The precursor is already stored in SNF.

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

   - The excess precursor is already dealt with by SNF staff.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: